

Chapter

Aircraft Health Monitoring Systems (AHMS) Applications in Aviation

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Abstract

Aviation holds a significant position in the development and application of health monitoring technologies. In recent years, the integration of Advanced Health Monitoring Systems (AHMS) in aircraft has become increasingly important for ensuring flight safety, operational efficiency, and cost-effective maintenance. This chapter provides an overview of the historical development of AHMS in the aviation sector and discusses the fundamental approaches adopted in aircraft maintenance practices. Particular emphasis is placed on the measurement techniques and sensor systems utilized to monitor the health status of critical components. Furthermore, the chapter explores various data processing and modeling techniques used to analyze the information collected from onboard sensors, including predictive algorithms and condition-based maintenance strategies. By reviewing these technologies and methods, the chapter aims to highlight the growing role of health monitoring in enhancing aircraft reliability, reducing maintenance costs, and supporting proactive decision-making within the aviation industry.

Keywords: structural health monitoring, predictive maintenance, prognostic, aircraft maintenance, digital twin

1. Introduction

Aircraft Health Monitoring Systems (AHMS) have become a critical technology to improve the safety of aircraft, optimize maintenance costs and maximize operational efficiency. AHMS enable real-time monitoring, damage detection and life prediction of aircraft components (engines, wings, fuselage, avionics). It is also very useful tool for the development of preventive maintenance strategies [1]. Another important objective of AHMS is to detect these small changes before failure occurs at the earliest possible opportunity, thereby minimizing downtime, operating costs and maintenance costs, and taking corrective action to reduce the risk of catastrophic failure, injury and even loss of life [2].

In recent years, AHMS technologies have undergone a major transformation with advanced sensor networks, machine learning algorithms and digital twin

applications. Fiber optic sensors, piezoelectric materials and wireless data transmission have significantly increased the sensitivity and usefulness of AHMS [3]. In addition to improving flight safety, the adoption of SHM in the aviation industry has resulted in significant savings in operating costs by reducing planned maintenance processes, reducing flight postponement rates due to unplanned failures, and preventing further damage by taking corrective action before parts fail [4].

Typical applications of AHMS in aviation include engine performance monitoring, structural crack detection, composite material damage analysis and health assessment of avionics systems [5]. In addition, AHMS, integrated with digital twin technologies, have added a new dimension to failure prediction and maintenance planning by simulating virtual models of aircraft [6].

2. Health monitoring: Definition, history and uses

Briefly, Aircraft Health Monitoring Systems (AHMS) are advanced technology solutions that monitor the condition of structural and mechanical components of aircraft in real time or periodically. By analyzing data collected through sensors, these systems aim to detect potential failures in advance, improve flight safety and optimize maintenance costs.

2.1 The history and important concepts of aircraft health monitoring

Pioneering fields, such as aviation and aerospace, have played a major role in the development of structural health monitoring techniques to the current level, with the main motivating factors such as ensuring flight safety and reducing maintenance costs.

Health monitoring applications in the aerospace industry, as in other areas of technology, have evolved and matured over time with new needs and developing technology. Understanding time-dependent damage progression can be considered as the first important step in structural health monitoring applications. Engineers and academics began to recognize the impact of cyclic stress on the lifespan of components. This understanding led to the development of the S-N curve, which relates the magnitude of cyclic stress to the logarithm of the number of cycles to failure [7]. Initially, the focus was on the safe life approach, which focuses on designing structures to last for a predetermined period without failure. This philosophy manifested itself in structures subjected to repetitive loads, such as railroad bridges and steam boilers, during the steam engine era in the mid-1800s [8, 9].

Another very important building block on the road to AHMS applications is the damage tolerant design philosophy. The evolution of the damage tolerant design philosophy dates back to the mid-twentieth century and marks a significant shift in aerospace engineering practices.

By the 1970s, the limitations of the safe life approach led to the emergence of the damage tolerance philosophy. This new paradigm was introduced to account for the potential growth of existing cracks and defects in materials, rather than focusing solely on preventing their occurrence. Regulatory requirements for damage tolerance analysis were formally integrated into aviation documentation in 1976 and have significantly shaped design standards for aircraft and other heavily loaded aerospace structures [9, 10]. Historical events such as the Antonov An-10 aircraft crash have highlighted the need for robust design methodologies that can accommodate

undetected damage and have led to the creation of damage tolerant structures [10]. This philosophy not only considers the initial integrity of a structure but also emphasizes the importance of monitoring and managing the growth of any damage that may occur during its operational life.

The concept of health monitoring in the aviation industry has evolved significantly, driven by the continuous pursuit of improved safety and operational efficiency. In the 1970s, pioneering airlines such as British Airways, Air France, and TAP Air Portugal launched the first Flight Data Monitoring (FDM) programs. These programs aim to analyze flight data to enhance operational safety and identify performance-related issues [11].

This early groundwork laid the foundation for contemporary Prognostic Health Management (PHM) systems, reflecting a long-standing commitment to proactive safety measures. As industry progressed, the development of the Aircraft Communications Addressing and Reporting System (ACARS) in the 1980s marked a transformative moment in aircraft monitoring. This automated data link communication system enables the seamless transmission of critical data from the aircraft to airline operations centers, eliminating the need for manual intervention by mechanics [12].

In the early 1990s, these monitoring systems began to gain wider acceptance, reflecting industry-wide recognition of the need for proactive safety measures. Today, the integration of advanced sensor technologies and artificial intelligence into health monitoring systems continues to shape the future of aviation, providing major benefits in the areas of early fault detection and predictive maintenance [13, 14].

Table 1 shows several important events and studies that have played a milestone role in the development of structured health monitoring technology to date.

Health monitoring systems used in aircraft and other aerial platforms can be classified as On-board Health Monitoring Systems (OHMS) and ground-based health monitoring systems (GHMS). Aircraft Health Monitoring Systems (AHMS) utilize a network of sensors installed on critical aircraft components and continuously monitor parameters such as engine performance, structural integrity, and various system functions. The collected data is transmitted to GHMS, where ground maintenance teams analyze the information to make informed maintenance decisions. This system enables timely interventions that prevent costly repairs and reduce in-flight failures.

On-board Health Management Systems (OHMS) are critical components in modern aviation systems designed to ensure aircraft safety, reliability, and operational efficiency. These systems employ advanced sensors, data analytics, and predictive technologies to monitor the health of various aircraft systems in real time, enabling proactive maintenance and reducing lifecycle costs [27, 28].

OHMS leverages a network of integrated sensors that continuously collect and analyze operational data from the aircraft during flight. This capability allows maintenance teams to detect potential issues before they evolve into failures, thereby minimizing delays and enhancing passenger safety and comfort [29].

One significant advantage of OHMS is its ability to perform real-time monitoring of aircraft health. Sensor data is instantly transmitted to ground stations or cloud-based platforms for immediate analysis, enabling timely interventions [30]. In other words, OHMS simultaneously generates the database required by GHMS.

The benefits of OHMS include providing a holistic approach to aircraft maintenance by integrating data from health monitoring systems with maintenance databases and operational programs. Automated systems within OHMS can issue maintenance alerts and safety warnings based on predictive analyses that help inform

Year	Milestone	Institution/ Key contributors	Significance
1860	Metal fatigue theory formulation	August Wöhler	Basis for damage tolerance principles [14].
1921	Vibration analysis fundamentals	Stephen Timoshenko	Enabled dynamic structural analysis [15].
1946	Ultrasonic flaw detection	Firestone	First practical ultrasonic NDT method [16].
1950	Acoustic emission discovery	Josef Kaiser	Real-time crack detection [17].
1955	Experimental modal analysis	Bishop & Gladwell	Vibration-based damage ID [18].
1958	USAF Aircraft Structural Integrity Program (ASIP) was established	United States Air Force	Safe life and damage tolerant design concepts were born [19].
1970	The concept of digital twin was first used	NASA	The concept of digital twin was first used during Apollo 13 mission [20].
1978	Fiber Bragg grating sensors	Butter & Hocker	Distributed strain measurement [21].
1970–1980	Thermographic inspection using Focal Plane Array	IBM	First practical two-dimensional arrays of infrared detectors to be produced and effectively cooled [22].
1991	The Digital Twin (DT) concept emerged	David Gelernter	The starting era of digital twin [23].
1995	Embedded piezoelectric sensors	Chang F.K., Stanford University	Embedding arrays of piezoelectric sensors within composite laminates during manufacturing [24].
2012	Vehicle Health Management Project	NASA	One of the earliest implementations, when NASA developed prototype digital twins for aircraft engines and structural components to predict component failures before they occurred [25].
2013	ARP6461 Guidelines for Implementation of Structural Health Monitoring on Fixed Wing Aircraft	SAE International	Important industry-wide recommended practice documents specifically for aviation SHM implementation [26].

Table 1.
Key milestones in the development of aircraft health monitoring technology.

both in-flight decisions and ground operations [31]. This consequently reduces aircraft downtime and associated costs [32, 33].

Ground-based health management systems (GHMS) are essential components in modern aviation, providing critical analysis and monitoring capabilities for aircraft operating environments and conducting in-depth analyses of aircraft component health. These systems operate outside the aircraft and utilize advanced data analytics to enhance safety and efficiency throughout the aircraft's lifecycle. Not being constrained by aircraft weight and volume limitations, GHMS systems benefit from

powerful data analysis hardware and servers. More complex operations such as health information of the entire fleet, maintenance information, and calculation of efficient flight routes for passenger aircraft are performed within GHMS. It enables maintenance teams to monitor fleet-wide performance and predict potential failures before they occur, reducing unplanned maintenance and operational costs [34, 35].

GHMS collects data from multiple aircraft within a fleet and processes them at maintenance, repair, and overhaul (MRO) centers. This data includes vital indicators such as vibration, temperature, and structural integrity, which are necessary for assessing aircraft health [31].

By applying advanced data analytics techniques, OHMS can identify patterns and anomalies indicating wear or impending failures, allowing for timely maintenance interventions [34, 35]. The transition from reactive maintenance to predictive maintenance is a fundamental aspect of GHMS. These systems analyze historical and current data using complex data analytics to predict potential failures [30, 35]. This proactive approach significantly increases aircraft availability and operational efficiency while minimizing maintenance costs. The success of predictive maintenance largely depends on the quality of the analyzed data, making data integrity a top priority [35, 36].

2.2 Basic approaches used in aircraft health monitoring Modeling

In aircraft health monitoring applications, various models are utilized to make accurate maintenance decisions, avoid unnecessary maintenance costs, and ensure flight safety. The first models were mostly physics-based and began to be used in the 1970s and 1980s. These models were developed based on the principles of materials science, structural mechanics, and fatigue theories [37].

Data-based models became widespread in the late 1990s and especially in the early 2000s. In this period, with the development of sensor technologies, increased data storage capacities and the advancement of machine learning algorithms, it has become possible to extract meaningful maintenance information from large data sets [38]. After the 2010s, hybrid models combining both approaches have gained importance [38, 39].

2.2.1 Physics-based modeling prognostic approach

Physics-based models work with mathematical equations based on the material properties, geometry and dynamic behavior of the structure. By understanding the underlying physics of what happens, engineers can predict potential failures and maintenance needs before they occur, improving safety and efficiency in operations [39–41]. Finite element analysis (FEA) and modal analysis are the most commonly used tools in this approach [1].

Its advantages include the ability to operate with limited data, explain fusional processes, and model long-term material behavior. However, it also has disadvantages, such as requiring high computational power and facing difficulties in modeling complex systems, where parameter uncertainties—such as material properties and temperature conditions—can lead to significant errors.

2.2.2 Data-based modeling prognostic approach

Data-based modeling focuses on leveraging historical data collected from aircraft systems to identify patterns and anomalies [39–41]. The data-driven model detects

anomalies or damage by analyzing sensor data (accelerometer, strain gauge, acoustic emission, etc.) obtained from the structure with machine learning (ML) and statistical methods. These models are based on the detection of irregularities in the data instead of physical equations. This method provides real-time insights and predictive analytics using algorithms that learn from existing data. It enables proactive maintenance before potential damage occurs. This results in lower failure rates, shorter maintenance times and safer operation. Data-based models are particularly effective in scenarios where physical models may be too complex or costly to develop [41]. Its advantages are that it does not require complex physical models, provides highly accurate results in case of large data sets, and is suitable for real-time monitoring. The disadvantages are that it requires high quality and abundant training data, the difficulty of physical explanation of the results due to its “black box” structure, and the limited extension of the model to different structures.

2.2.3 Hybrid approach

Hybrid SHM models combine the advantages of data-driven and physics-based approaches to provide more reliable and interpretable results. For example, hybrid models can use data to refine and calibrate physical models, improving their predictive capabilities while at the same time ensuring they are grounded in the reality of observed performance [39, 40]. These models are becoming increasingly common, especially in high-risk areas such as aviation, energy and defense industries. The advantages are high accuracy, less data requirement and interpretability. Disadvantages include the need for expertise, high computational power and the difficulty of compatibility of Physics and Data models.

2.2.4 Digital twin approach

A digital twin is a virtual copy of a physical asset or system (airplane, bridge, wind turbine, etc.) fed with real-time data. Digital twins used in AHMS simulate the current state of the structure, enabling failure prediction, optimization and development of maintenance strategies.

In the aviation industry, digital twins are used to improve maintenance processes and increase the safety and efficiency of operations. By enabling the simulation of various scenarios, they help predict wear and tear on components and facilitate proactive maintenance strategies.

By integrating digital twin technology into avionics processes, manufacturers such as Boeing have underscored its importance in modern aviation management. As technology continues to evolve, the potential for digital twins to create comprehensive models spanning entire fleets or organizations will further revolutionize the aviation industry, providing real-time visibility and operational insights on a much larger scale [42–45]. The basic components of a digital twin are the physical system (real structure equipped with sensors), the virtual model, data communication and algorithms that generate predictions.

The advantages are real-time monitoring capability, failure prediction, cost savings, and no “black box” problem thanks to physical models. The disadvantages are high computational power, data security risks and difficulty of integration.

2.3 Measurement techniques used in aircraft health monitoring systems

For a typical AHM system to function properly, a sensor network that measures the required health data, a data acquisition and analysis unit that collects the data

from the sensors and analyzes the necessary data in real time, user interfaces designed for the pilot and maintenance team, and algorithms that analyze the data and report the current status and future maintenance recommendations.

In order for the AHM system to fulfill these tasks, it is first necessary to know the stresses and environmental conditions to which the aircraft is exposed during its use. For this purpose, a large number of different purpose-built sensors are used [20]. The main sensor types used in AHMS are given in **Table 2**.

2.4 Application areas of health monitoring systems in aircraft

Various factors, such as the advancement of aircraft designs, innovations in sensor technology, and the development of aircraft maintenance concepts over the years, have enabled the use of health monitoring applications in different parts of aircraft over the years. In this chapter, the main areas where AHMS is used are aircraft structures, gas turbines and accessory units, gearboxes, hydraulic equipment and landing gear.

Sensor type	Measurement characteristics	Typical application examples
Strain gauges	Measures and monitors strain on the object resulting from applied force or temperature	Airframe structural parts such as longerons, bulkheads
Load Cells	Used for measuring tension, compression, pressure or torque	Actuators
Accelerometer	Measures dynamic or static acceleration force in various directions	Airframe, landing gears
Temperature Sensors	Measures temperatures, usually by thermocouples, Infrared sensors or RTDs	Air inlet, compressors and turbines
Fiber Optic Sensors	Measures stress, strains or temperature	Airframe structural parts, cylinders
Acoustic Emission Sensor	Measures the current state of materials under stress. Used to dynamically detect the growth of cracks or corrosion	Airframe structural parts
Vibration Meters	Measures high or low frequency vibrations of the structure	Gear boxes, turbine casings
Linear Variable Differential Transformer (LVDT)	Measures the linear position of the object	Flap actuators
Tiltmeter	Measures the inclined angle of the object	Determination of aircraft, airframe structural parts
G - level Indicator	Measure gravitational forces	Landing gears
Pressure Transducers		Compressors, fans, air inlet pressures, fluid lines
Proximity Sensors	Measures the distance of the objects	Determining turbine shafts, gear shafts revolution speed
Piezoceramic sensors	Measures strain and vibrations	Embedded in structural composite wing panels

Table 2.
Different types of sensors used in AHMS for data collection and their example applications.

2.4.1 Aircraft structural parts health monitoring

The first examples of health monitoring applications in the aerospace industry were carried out on aircraft structural parts. The beginning of these applications can be traced back to the damage tolerance approach design philosophy of the US Air Force, which is also mentioned in history. In the damage tolerance design philosophy, a certain amount of damage is allowed to occur and progress throughout the lifetime of aircraft components to reduce their weight and maximize their performance. The speed at which these damages progress and their critical dimensions are determined at the design stage through calculations and tests. This makes it possible to work with parts that have the exact structural strength required by the design, rather than using heavy and over-engineered parts that are so heavy that they will never fail during their lifetime [19].

Aircraft components, such as spars, longerons and bulkheads, are critical components that provide the main structural integrity of the aircraft. These structural elements are vital in aircraft design as they enable the airframe to withstand various aerodynamic and operational stresses while maintaining the desired shape and function.

The main damages in aircraft structural components that are monitored using structural health monitoring techniques are fatigue cracks, corrosion and wear. Fatigue cracks are seen especially in areas of stress concentration, such as rivet holes, bolt thread bottoms and sharp corners on structural parts. Corrosion damage is generally observed in parts that are exposed to the atmospheric sun, moisture, salt water industrial atmosphere. Protective coatings and paints are used to prevent corrosion in these parts. The primary function of these protective layers is to act as a barrier, preventing the corrosive environment from interacting with the underlying metallic material.

When one thinks of Structural Health Monitoring (SHM), the Aircraft Structural Integrity Program (ASIP) comes to mind. Over the decades, the program has evolved significantly, adapting to the increasing complexity of modern aircraft designs and incorporating advanced methodologies such as damage tolerance and fracture control technologies [46, 47].

Today, ASIP uses advanced health monitoring applications that utilize real-time data to assess aircraft structural integrity, enabling proactive maintenance strategies that significantly improve aviation safety standards [46].

2.4.2 Gas turbine engine health monitoring

Gas turbine engines consist of several key components that work together to produce thrust and power. The primary parts of a gas turbine include the compressor, combustion chamber, turbine and in some cases a power turbine. These elements are essential for engine operation and efficiency [48].

Gas turbine engine monitoring techniques are critical to ensure the reliability and efficiency of engine operations. These methods primarily focus on detecting early signs of failures and assessing the overall health of the engine.

The development of Engine Health Monitoring Systems (EHMS) in aviation can be traced back to their first use on military helicopters in the late 1990s and early 2000s. These early systems laid the groundwork for the more sophisticated technologies that would follow. Over time, advances in computing power and forecasting algorithms have significantly improved the accuracy and efficiency of these monitoring systems, enabling a more precise collection of data based on past performance and operational use cases [49].

Aircraft Engine Health Monitoring Systems (EHMS) provide numerous benefits that significantly improve the efficiency and reliability of aviation operations.

One of the most important benefits of Aircraft Engine Health Monitoring Systems (EHMS) is their contribution to flight safety. These systems continuously monitor complex engine parameters, enabling early detection of potential problems before they become serious flight safety issues. This proactive approach increases confidence in safety practices among aviation operators and helps prevent unexpected downtime caused by engine-related issues, ultimately improving operational reliability [50, 51].

EHMS can reduce maintenance costs by up to 30% by helping to detect and prevent maintenance issues before they lead to failure [12, 52].

Integration of advanced technologies, including machine learning and historical data analysis, enables maintenance teams to accurately predict engine maintenance requirements and efficiently plan interventions, minimizing unscheduled maintenance events and increasing aircraft availability [53, 54].

These systems help aviation companies effectively manage operational risks. EHMS helps airlines comply with regulatory standards and extends the life of their aircraft fleet [55, 56].

AHMS adoption leads to an increase in overall operational efficiency. By optimizing maintenance strategies through advanced algorithms, airlines can reduce downtime and improve scheduling, which ultimately increases the efficiency of fleet operations [57, 58].

As a result, the aviation industry benefits from a comprehensive approach that not only enhances safety and reliability but also improves the economic viability of airline operations [54, 59].

Modern gas turbine health monitoring systems utilize data from a variety of engine-mounted sensors to provide early warnings of potentially dangerous conditions [60]. These systems can operate either in “on-line” mode, which analyzes in real time during the flight, or in “off-line” mode, which allows a comprehensive review of the recorded data after the operation [61].

The use of sensor data requires relatively few detailed theoretical system models, simplifying condition assessment and making it easier for engineers to effectively monitor and manage engine performance [60].

2.4.2.1 Temperature monitoring

Temperature monitoring is a vital aspect of gas turbine engine health assessment. For example, an excessive rise in turbine inlet temperature (TIT) or exhaust outlet temperature (EGT) could indicate permanent damage to the engine and a major failure that puts flight safety at risk. Methods for monitoring temperature typically involve the collection of multiple sensor readings that reflect the operating state of the motor [56, 59].

Given the high ambient temperature, especially for hot section temperature readings, special sensors must be used to accurately measure temperatures without being adversely affected by extreme conditions [62].

2.4.2.2 Vibration-based monitoring

Vibration measurements serve as a fundamental tool for global monitoring of the health status of a gas turbine. The principle underlying this technique is that the dynamics of a gas turbine system change when faults begin to occur, resulting in

altered vibration patterns compared to those observed in the healthy state [63, 64]. For example, a crack in the compressor blade or an imbalance caused by even a small part breakage is quickly detected by vibration sensors.

Recent developments show that gas turbine manufacturers are increasingly adopting data-driven vibration-based condition monitoring approaches that improve reliability and availability without relying on complex physics-based models that require numerous assumptions [63].

2.4.2.3 Oil analysis

Oil system monitoring has been one of the oldest and most established techniques in gas turbine engine monitoring. While it has a long history, advances in technology have led to improved methodologies that provide deeper insights into engine performance and condition [65]. For example, in the event of wear in the bearings on the engine, small metal particles mixed into the oil are detected, and further damage to the bearing or engine is prevented [61].

2.4.2.4 Gas turbine performance monitoring

Gas turbine performance monitoring serves as an important tool for early detection of failures. By developing models based on engine data, both linear and non-linear relationships specific to particular engines can be captured. This allows manufacturers to detect defects early during pass tests, significantly reducing the risk of complete component failure. For example, compressor compression ratio is a good performance metric, and a decrease in this value indicates damage to the compressor, such as wear, etc. [63].

Smart embedded sensing technologies further support these monitoring efforts, facilitating smooth operation and improved reliability [66].

2.4.2.5 Life consumption monitoring

Gas Turbine Engine Life Consumption Monitoring is an effective way to prevent in-use component failures. Historically, maintenance strategies were based on fixed time, but modern practices favor condition-based approaches that utilize life monitoring systems specific to each aircraft engine. For example, engine life under idle or after burner (AB) conditions is recorded on an aircraft-by-aircraft basis. By calculating the effect of each regime on engine life differently, the wear level of the engine of each aircraft in the fleet is calculated, and maintenance periods are determined accordingly. These systems have been particularly successful in extending the service life of aging fleets by automating the measurement of engine life [61, 67].

2.4.3 Accessories and auxiliary systems health monitoring

Health monitoring techniques for aircraft gearboxes, landing gear and hydraulic equipment are essential to ensure safety, reliability and operational efficiency in aviation. These components are essential for the proper functioning of various aircraft systems, with gearboxes enabling power transmission, landing gear ensuring safe take-off and landing, and hydraulic systems controlling critical functions. These systems are integral to various functions, including landing gear operation

and control surfaces, and rely on the efficient transmission of energy through high-pressure fluids. However, the complexity of hydraulic systems requires meticulous maintenance and monitoring to reduce the risks associated with potential failures.

The importance of effective health monitoring is manifested in its ability to detect potential failures early, thereby minimizing maintenance costs, reducing unplanned downtime and improving overall aircraft performance. Various techniques are used to assess the condition of accessories and auxiliary systems, leveraging advanced technologies for predictive maintenance.

2.4.3.1 Vibration analysis

Vibration analysis is a widely adopted technique for monitoring the operating condition of gearboxes. It involves measuring the vibration patterns produced by the gearbox during operation. When components are in good condition, they produce specific vibration patterns. However, as wear and tear occur on bearings and gears, these patterns change, which is a precursor to potential failures such as misalignment, imbalance or bearing failures [68, 69].

Vibration analysis is based on collecting vibration readings using sensors and then analyzing this data on established bases to identify anomalies. Serious failures in hydraulic units, such as gear breakage, bearing failure, etc., often manifest themselves in the form of sudden increases in vibration values.

2.4.3.2 Thermography

Thermography or thermal imaging uses infrared cameras or thermocouples through the fluid line to detect temperature variations inside gearboxes. Abnormal heat values often indicate problems such as excessive friction or mechanical wear [70, 71]. By monitoring temperature changes, technicians can diagnose potential problems before they lead to catastrophic failures.

2.4.3.3 Oil analysis

Oil analysis is another vital monitoring technique. Regular sampling and analysis of the oil used in gearboxes can reveal critical information about the condition of internal components. The analysis focuses on detecting contaminants, wear particles and oil quality degradation that could indicate problems such as bearing wear or inadequate lubrication [72].

This proactive approach ensures that maintenance is performed at optimal intervals, prolonging the life and safe operation of hydraulic components such as gearboxes, bearings and pumps.

2.4.3.4 Load and stress monitoring

Various methods have been developed to monitor the health of landing gears, often involving various sensors and collection systems. These methods enable the structural integrity of the landing gear to be analyzed in real time, allowing early detection of potential problems such as wear and tear [72, 73].

Common approaches include direct load measurements, strut detection and weight/balance calculations, which collectively contribute to a comprehensive understanding of the condition of the landing gear [74, 75].

Structural Health Monitoring (SHM) systems play a crucial role in assessing the condition of landing gear. These systems use advanced sensors integrated into the landing gear assembly to monitor loads and detect damage escalation in real time. Implementing SHM not only improves safety by facilitating condition-based maintenance (CBM) but also extends the service life of landing gear components [75].

By continuously monitoring the health status of the landing gear, potential failures can be detected at an early stage, and thus maintenance actions can be planned before failures occur [73, 76].

3. Conclusions

Maximizing flight safety has been an ongoing and never-ending demand in the aviation industry since its inception. To meet this demand, Aircraft Health Monitoring Systems (AHMSs) have been continuously evolving for many years. This historical development process not only emphasizes the importance of learning from past experiences but also demonstrates how technology has constantly evolved to meet the growing demands of air transportation and safety regulations.

Innovations in sensor technology and computational models have paved the way for powerful computing and health monitoring systems, such as digital twins. As we enter the era of big data, the integration of health monitoring systems into maintenance strategies is progressing at an accelerating pace.

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